

Experimental Measurements of the Flow and Heat Transfer of a Square Jet Impinging on an Array of Square Pin Fins

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ABSTRACT

An experimental investigation was conducted to explore the flow behavior, pressure drop, and heat transfer due to free air jet impingement on square in-line pin fin heat sinks (PFHS) mounted on a plane horizontal surface. A parametrically consistent set of aluminum heat sinks with fixed base dimension of 25 x 25 mm was used, with pin heights varying between 12.5 mm and 22.5 mm, and fin thickness between 1.5 mm and 2.5 mm. A 6:1 contracting nozzle having a square outlet cross sectional area of 25 x 25 mm was used to blow air at ambient temperature on the top of the heat sinks with velocities varying from 2 to 20 m/s. The ratio of the gap between the jet exit and the pin tips to the pin height, the so-called tip clearance ratio, was varied from 0 (no tip clearance) to 1. The stagnation pressure recovered at the center of the heat sink was higher for tall pins than short pins. The pressure loss coefficient showed a little dependence on Re , increased with increasing pin density, and pin diameter, and decreased with increasing pin height and clearance ratio. The overall base-to-ambient thermal resistance decreased with increasing Re number, pin density and pin diameter. Surprisingly, the dependence of the thermal resistance on the pin height and clearance ratio was shown to be mild at low Re , and to vanish at high Re number.

KEYWORDS

pin fin heat sinks, impinging jet, electronics cooling.

NOMENCLATURE

A	area [m^2]
a	fin side [m]
Bi	Biot number
b	distance between two fins [m]
CL	clearance ratio [h/H]
C_p	pressure loss coefficient
D	distance along the diagonal of the heat sink measured from its center [m]
H	pin height [m]
h	distance between the jet exit and the pin tips [m]
h^*	heat transfer coefficient [$W/m^2 \cdot K$]
k	thermal conductivity of the fin material [$W/m \cdot K$]
N	total number of pins in the heat sink
n	number of rows of pin in the heat sink
P	pressure [Pa]
P^*	fin cross sectional perimeter [m]
PR	ratio of static pressure to total pressure
Q	heat transfer rate [W]
R	overall heat sink thermal resistance [K/W], $\frac{\Delta T_B}{Q_B}$
R'	ratio of R over R at $h/H=0$
Re	Reynolds number, $\frac{V.W}{u}$
T	temperature [K]
V	velocity at the exit of the nozzle [m/s]
W	heat sink width [m]

X distance along the centerline of the heat sink measured from its center [m]

Greek symbols:

ΔP pressure drop [Pa]
 ΔT_B temperature difference between the heat sink base and ambient temperature [K]
 ΔT_H temperature difference between the heater and ambient temperature [K]
 ρ jet air density [kg/m³]
 d uncertainty
 μ kinematic viscosity [m²/s]
 η efficiency

Subscripts:

$A/D_{\text{converter}}$ analog to digital converter
 B heat sink base
 c fin cross section
 DR digital readout
 ex exit of the control volume described in Figure 1.
 ΔT_B temperature difference between the heat sink base and ambient
 $exit$ at the nozzle exit
 f fin
 H heater
 in inlet of the control volume described in Figure 1.
 $inlet$ at the nozzle inlet
 L lost
 max maximum
 o overall
 P pressure
 P_T total pressure
 P_{exit} pressure at the exit of the nozzle
 P_{inlet} pressure at the inlet of nozzle
 PT pressure transducer
 Q_T total heat supplied by the heater
 Q_L heat lost from the heat sink by conduction and radiation
 s static
 T total
 ∞ ambient

INTRODUCTION

In order to extend the usefulness of direct air cooling in the thermal management of high power density electronic packages, the most economical and simplest approach is to use extended surface heat sinks. Heat sinks are most commonly operated in a configuration in which the flow enters and exits from

the sides [1-3], the so-called side-inlet, side-exit (SISE) configuration. In this configuration, both parallel plate and pin fin type extended surfaces have been studied. Parallel plate arrangements have been exhaustively researched, and ample information exists to adequately design and simulate optimized configurations. Pin fin heat sinks have been more resistant to modeling because the flow in such geometries is dominated by three-dimensional features such as flow separation and endwall vortices at the pin-base junction. Hence models are more dependent on experimentally derived correlations, which are sparse in the literature.

Shaukatullah et al. [1] studied the design and optimization of parallel plate and in-line square pin fin heat sinks for different fin height, thickness, and spacing under a variable cross flow angle, at velocities under 5m/s. Dogruoz et al. [2] presented an experimental and modeling study on the hydraulics of a set of in-line square pin fin heat sinks under cross-flow in a channel, allowing top clearance. A simple 1-D bypass model was developed to predict the pressure drop, which was then compared with the experimentally acquired one. It was shown that the inlet and exit losses are as important as the core effects. Maudgal et al. [3] showed data for convection from three different in-line pin fin arrays under various fluid flow angles, fin tip clearance, and flow rates.

As package power density has increased, system manufacturers have found it necessary to deliver higher velocity flow directly to the heat sink, in order to further decrease the base to ambient thermal resistance. In this pursuit, the positioning on dedicated small fans blowing directly downward onto a heat sink has been the most common manifestation of this principle.

The use of jet impingement for spot cooling has been studied mostly for jets blowing onto plane surfaces. Martin [5] has reviewed the literature for single and multiple jets, in confined and unconfined configurations. The study of impinging jets blowing onto discretely heated surfaces, such as on to electronic packages, is relatively recent [6,7]. Amano [6] numerically characterized the flow and heat transfer from a flat plate with a constant heat source under a normally impinging axisymmetric turbulent jet. He also showed good agreement with previously obtained experimental data. Guarino and Manno [7] showed a simulation of a laminar air jet impingement within a geometry representing a laptop computer, where a finite control volume technique was adopted to solve for the velocity and temperature fields.

The impingement of a jet onto the top of a heat sink, the so-called Top inlet – Side exit (TISE) configuration has seen very little fundamental study,

and accurate data that can be used for design optimization and CFD validation is highly restricted [9,10,11]. Jung and Maveety [9] presented a numerical simulation using a CFD package in modeling an air jet impinging from a round nozzle with various velocities over three different heat sink geometries. Brignoni and Garimella [10] presented experimental optimization data for a given pin-fin heat sink under a confined air jet impingement at a fixed nozzle-to-target spacing, using a variety of nozzle configurations at different flow rates. They reported average heat transfer coefficients and thermal resistances as a function of Reynolds number. El-Sheikh and Garimella [11] presented correlations for the Nusselt number for confined air jet impingement on different pin fin heat sinks at various nozzle diameters and Reynolds numbers. In both studies [10,11] the nozzle diameter was generally less than the width of the heat sink.

This paper presents experimental results for a well-defined TISE pin fin configuration, illustrated in Figure 1. The problem to be studied is that of a pin fin heat sink with a square base, $W \times W$, that is cooled by flow from a square nozzle of the same $W \times W$ cross-section, impinging onto the top of the heat sink. The flow is conditioned so that it approaches the heat sink with uniform velocity, with no swirl, and with low free stream turbulence. We will present results for the hydraulic and thermal behavior of heat sinks with square pins, over a wide variation of pin height, pitch and thickness, both with and without tip clearance. The objective of this paper is to clarify the physics of the flow and heat transfer in this generic geometry, as a step towards developing better methodologies for optimal design of this advanced air cooling technology. To our knowledge, this is the first systematic experimental study for this fundamental problem.

EXPERIMENTAL APPARATUS AND PROCEDURE

I. Flow Measurements

The experimental apparatus is shown in Figure 2. A 6:1 contracting nozzle having inlet and exit cross-sectional areas of 150mm x 150mm and 25mm x 25mm, respectively, was constructed with 3.75mm smooth fiberglass walls and used to create a square jet impinging on the top of the heat sinks. The blower provides flow to a 1.75m long diffuser that exits into a 40 mm thick layer of honeycomb followed by three mesh stainless steel wire screens. Two pressure taps were used to measure the static pressure at the inlet (150mm x 150mm) and at the outlet (25mm x 25mm) of the nozzle. These pressures were used to calculate

the area-averaged velocity V at the exit of the nozzle. The heat sinks were placed on a movable table underneath the nozzle. This design allows a variable clearance ratio h/H , which is the ratio of the gap between the jet exit and the pin tips, to the pin height. Four aluminum rods, fixed from both ends by two horizontal rigid frames, are used to vertically slide a 450mm x 450mm horizontal aluminum frame by adjustment of a manual jack attached below it. A removable 300mm x 300mm polished aluminum plate with a cutout for the heat sink is placed in the frame. A 25mm x 25mm cutout, 2.5mm deep, was made in the middle of this plate under the outlet of the nozzle so that the base of the heat sink is flush with the polished surface as shown in Figure 3a. This test surface is equipped with 16 pressure taps distributed as shown in Figure 3b, 8 along the centerline and the rest along the diagonal of the square base. Pressure taps were also placed in the base of the heat sinks in various locations as shown in Figures 3a and 4; the positions of these taps are dependent on the geometry of each heat sink. A speed controllable centrifugal blower, which is connected to the top of the diffuser with a duct, provides precise flow to the jet. The jet impinges with various velocities at several clearance ratios on the heat sink. In this study, twenty aluminum pin fin heat sinks are investigated as described in Table 1. These heat sinks, which are a subset of a set used by Shaukatullah et al [1], were also studied by Dogruoz et al. [2] to determine pressure drop and heat transfer under SISE conditions.

Procedure

A type 120A Baratron high accuracy differential pressure transducer was used to measure the time-averaged static pressures at the different locations. The Baratron transducer yields a linear, high level DC output signal that is proportional to the pressure. The signal is sent to a MKS Type 660B digital readout, then to the computer through a National Instruments AT-MIO-16A-2 A/D converter. The pressure is sampled for six seconds at a 1000 Hz sampling rate, then averaged. After changing to a new measurement tap, the initiation of measurements is delayed for six seconds in order to account for transient effects.

The velocity exiting the nozzle is calculated by applying the Bernoulli Equation between the inlet and exit of the nozzle, assuming the flow is inviscid, that is:

$$V = \sqrt{\frac{2(P_{inlet} - P_{exit})}{\rho}} \quad (1)$$

The static pressure at the inlet of the nozzle, P_{inlet} is taken to be equal to the total pressure, since the velocity of the flow is negligible at this location compared to the nozzle exit velocity V .

An overall pressure loss coefficient is defined as:

$$C_p = \frac{P_{Tinlet} - P_\infty}{\frac{1}{2} \rho V^2} \quad (2)$$

The behavior of C_p , in addition to the local static pressure mapping along the heat sink and test surface, is examined for the different heat sinks under the various flow conditions.

Uncertainty in pressure and related parameters

The uncertainty in the pressure measurement is estimated by propagating the various uncertainties using the method recommended by Moffat [4]. The different sources of uncertainties are shown in Table 2. The absolute uncertainty in the pressure reading is governed by the accuracy of the pressure transducer, the accuracy of the digital readout and the resolution of the A/D conversion board. The following relation is used to determine the absolute uncertainty of pressure:

$$(dP)^2 = (dP_{PT})^2 + (dP_{DR})^2 + (dP_{A/D.converter})^2 \quad (3)$$

which, after replacing the values given in Table 2, reduces to:

$$dP = \sqrt{(0.105942)^2 + (0.0013P)^2} \quad (4)$$

Relative uncertainty of the pressure dP/P tends to 0.13% as the measured pressure increases.

The absolute uncertainty in the velocity derived from Eq. 1 is given by:

$$dV = \sqrt{\left(\frac{\partial V}{\partial P_{inlet}} \cdot dP_{inlet}\right)^2 + \left(\frac{\partial V}{\partial P_{exit}} \cdot dP_{exit}\right)^2 + \left(\frac{\partial V}{\partial r} \cdot dr\right)^2} \quad (5)$$

Therefore, the relative uncertainty in the velocity reduces to:

$$\frac{dV}{V} = \sqrt{\left[\frac{dP_{inlet}}{2 \cdot (P_{inlet} - P_{exit})}\right]^2 + \left[\frac{dP_{exit}}{2 \cdot (P_{inlet} - P_{exit})}\right]^2 + \left[\frac{dr}{2 \cdot r}\right]^2} \quad (6)$$

As expected, dV/V decreases with increasing velocity, and varies between 3.7% for low velocities ($V=2\text{m/s}$) to 0.5% for high velocities ($V=20\text{m/s}$).

Similarly, the absolute uncertainty in C_p is given by:

$$dC_p = \sqrt{\left[\left(\frac{P_{exit}}{(P_T - P_{exit})^2}\right) dP_T\right]^2 + \left[\left(\frac{P_T}{(P_T - P_{exit})^2}\right) dP_{exit}\right]^2} \quad (7)$$

and its relative uncertainty is expressed by:

$$\frac{dC_p}{C_p} = \sqrt{\left[\left(\frac{P_{exit}}{P_T (P_T - P_{exit})}\right) dP_T\right]^2 + \left[\left(\frac{1}{(P_T - P_{exit})}\right) dP_{exit}\right]^2} \quad (8)$$

The relative uncertainty in C_p varies between 5.8% for low jet velocity ($V=2\text{m/s}$) to less than 0.05% at high velocity.

II. Heat transfer measurements

The same experimental setup, with a different test plate, was utilized in acquiring heat transfer data. A heat transfer surface, shown in Figure 5, was designed for this purpose. This surface is a 300mm x 300mm Plexiglas plate that has a 125mm x 125mm cutout in its center to seat a low conductivity block. The block consists of a low-density urethane foam (Renshape 5120) with dimensions 125mm x 125mm x 46.88mm. The top surface of the block is covered by a 3.12mm thick Phenolic plate that provides a smooth surface to the impinging flow. The Phenolic plate is 125mm x 125mm and has a 25mm x 25mm cutout in its center so that the heat sink base is flush with its top surface. A 25mm x 25mm, Kapton-encapsulated nichrome serpentine heater is placed in the middle of the upper surface of the Renshape block. The heater is adhered to a 25mm x 25mm x 0.62mm thick copper heat spreader. A very thin layer of thermal grease is placed between the heat spreader and the heat sink base in order to reduce thermal resistance. A number of thermocouples are utilized in order to measure various temperatures. The first one is placed underneath the heater to determine its maximum temperature. The second one is embedded in the heat sink base. Three thermocouples are placed at the nozzle inlet. One, type T, measures the temperature of the air jet referenced to an ice bath. The other two, type T and K, are used as references for the heat sink and the heater thermocouples, respectively. The heat sinks instrumented for heat transfer measurements were separate but nearly identical in geometry and finish as those used for pressure mapping.

Procedure

The thermocouples were used to measure the air jet temperature T_∞ , the temperature difference between the heat sink base and air flow $\Delta T_B, (T_B - T_\infty)$, and the temperature difference between the heater and air jet $\Delta T_H, (T_H - T_\infty)$. The heater and heat sink base temperatures were referenced to the jet temperature in order to reduce the uncertainty in the measurement. The induced DC voltage from the thermocouples is proportional to the temperature and is measured with a Fluke Hydra Model 2620 interfaced to the computer via a GPIB board. The signal is converted to temperature using a calibration table found by calibrating sample thermocouples against a NIST traceable resistance thermometer. Temperatures are not recorded until after steady state conditions are achieved, that is when no variation in the signal is observed for two consecutive minutes. The heat convected through the heat sink, Q_B is given by:

$$Q_B = Q_T - Q_L \quad (9)$$

A significant amount of heat is conducted into the block, and since the downward losses were not directly measured, they were modeled using a finite difference model of the system using a commercial conduction code [12]. The code was used to solve for the heat loss into the block by applying the following boundary conditions to the block surfaces: natural convection heat transfer to the bottom surface and the sides, and the heat transfer coefficient for a jet impingement given by Martin [5] on the top surface. The conduction heat losses varied from 9.54% to 19.87% of the total heat supplied, Q_T . The radiation heat loss from the heat sink to the atmosphere was accounted for by a rigorous gray diffuse surface model, and was predicted to be between 1.06% and 6.12% of the total heat supplied Q_T . Further details can be found in [13].

The heat sink base-to-ambient thermal resistance was the principle thermal metric characterized in the study and is defined as:

$$R = \frac{T_B - T_\infty}{Q_B} \quad (10)$$

Local measurements of temperature and heat transfer coefficient were not attempted in this initial experiment phase.

Uncertainty in temperature and related parameters

From the definition of the base-to-ambient thermal resistance given in Eq. 10, it is obvious that the main uncertainty is in the uncertainty of the temperature difference between the heat sink base and the air jet.

In the measurement of ΔT_B , a type T thermocouple embedded in the heat sink base was directly referenced to a thermocouple in the air stream, in order to reduce the uncertainty in the measurement of ΔT_B by half.

The Least-Squares Regression Analysis for a single variable was utilized to determine the uncertainty in ΔT_B . A first order polynomial fit of the calibration temperature data was used, as suggested in [14]. The uncertainty in ΔT_B is found to be $d\Delta T_B = \pm 0.49^\circ\text{C}$.

The uncertainty in the overall thermal resistance R is governed by three main factors: the uncertainties in total heat transfer, losses, and temperature difference between the heat sink base and ambient. The following relation is used to predict dR :

$$dR = \frac{\sqrt{\left(\frac{\partial R}{\partial Q_T} \cdot dQ_T\right)^2 + \left(\frac{\partial R}{\partial Q_L} \cdot dQ_L\right)^2 + \left(\frac{\partial R}{\partial \Delta T_B} \cdot d\Delta T_B\right)^2}}{\quad} \quad (11)$$

The relative uncertainty in R , dR/R is:

$$\frac{dR}{R} = \frac{\sqrt{\left(\frac{dQ_T}{Q_T - Q_L}\right)^2 + \left(\frac{dQ_L}{Q_T - Q_L}\right)^2 + \left(\frac{d\Delta T_B}{\Delta T_B}\right)^2}}{\quad} \quad (12)$$

The relative uncertainty in Q_T is given by the power supply manufacturer as $dQ_T/Q_T = 0.5\%$. The uncertainties in the losses, dQ_L were derived rigorously and presented by Issa [13]. The resulting overall relative uncertainty in the resistance, dR/R , was found to vary from 2.89% to 7.23%.

RESULTS AND DISCUSSION

Fluid mechanics

For the heat sinks described in Table 1, experiments were conducted in order to measure local static pressure distributions and overall pressure drop. Air jet velocities of 2, 4, 5, 10 and 20 m/s were investigated for clearance ratios of $h/H = 0, 1/8, 1/4$ and 1. The behavior of the local static pressure distribution and the pressure loss coefficient C_p were examined with respect to Reynolds number, clearance ratio, pin height, pin thickness, and number of pins. Only cases for which the jet width, W was equal to the heat sink width were examined. Before studying the data, it will be instructive to anticipate some behaviors that appear to be obvious. Referring

to Fig. 1, the overall pressure drop, at minimum, is composed of the following components, some viscous and some inviscid:

- (1) Inviscid pressure change due to acceleration or deceleration of flow from the jet area to the exit area of the jet after it has turned (ΔP_{area})
- (2) Inviscid pressure drop due to acceleration of flow due to area reduction and viscous losses at the inlet of the heat sink (ΔP_{in})
- (3) Pressure drop in core of the heat sink due to viscous friction, flow separation, and shear layer mixing (ΔP_{core})
- (4) Inviscid pressure recovery due to area change at exit and viscous losses at the exit of the heat sink (ΔP_{ex})

Hence, we can write:

$$\Delta P_{Total} = \Delta P_{area} + \Delta P_{in} + \Delta P_{core} + \Delta P_{ex} \quad (13)$$

We note that except for ΔP_{area} , the loss mechanisms are the same as in parallel flow through the heat sink. The inviscid pressure change denoted by ΔP_{area} is due to the fact that the jet area is generally less than the cross-sectional area of the jet after it has turned 90° and become a wall jet. The deceleration leads to a pressure recovery. This pressure recovery can be estimated by the following analysis, for the case of zero bypass, $h/H = 0$. From Fig. 1, consider that the flow approaches the heat sink area, A_{in} , at an average velocity, $\bar{V}_{in}$, and exits with velocity, $\bar{V}_{ex}$, through the area A_{ex} . A_{in} and A_{ex} are the projected areas, not the flow area between pins. From the mass conservation relation, the following can be written:

$$V_{ex}^2 = V_{in}^2 \left(\frac{A_{in}}{A_{ex}} \right)^2 \quad (14)$$

if the fluid is incompressible.

Bernoulli equation between inlet and exit yields:

$$\Delta P_{area} = P_{in} - P_{ex} = \frac{1}{2} \rho (V_{ex}^2 - V_{in}^2) \quad (15)$$

Substituting (14) into (15),

$$\Delta P_{area} = P_{in} - P_{ex} = \frac{1}{2} \rho V_{in}^2 \left(\left(\frac{A_{in}}{A_{ex}} \right)^2 - 1 \right) \quad (16)$$

Therefore,

$$C_{p_{area}} = 1 + \frac{P_{in} - P_{ex}}{\frac{1}{2} \rho V_{in}^2} = \left(\frac{A_{in}}{A_{ex}} \right)^2 \quad (17)$$

Hence, by the geometry,

$$\frac{A_{in}}{A_{ex}} = \left(\frac{W}{4H} \right) \quad (18)$$

Table 3 lists the predicted $C_{p_{area}}$ and as expected shows that in all cases, deceleration of the jet as it spreads radially after clearing the heat sink leads to a strong pressure recovery.

The inlet loss, ΔP_{in} , and exit loss, ΔP_{ex} , in principle could be estimated using the approach described in Dogruoz et al [2], however, the loss coefficients, are unknown and not available in the literature. At the exit, prediction of losses is further complicated by the fact that the flow does not necessarily exit with a velocity vector normal to the heat sink exit plane. Prediction of the core pressure drop, ΔP_{core} , is complex because the flow turns 90° within the heat sink, hence the pressure drop coefficients for flow either parallel to or normal to the pins do not apply.

In this preliminary study, the goal was to observe the overall parametric behavior of the heat sinks with tip clearance. In order to distinguish and understand individual loss mechanisms, more detailed measurements of pressure and velocity fields are required. These detailed studies are underway. By Eq. (17), we expect $C_p \rightarrow 1$ as tip clearance increases or the pin density decreases. On the other hand, as pin density increases, C_p will increase. The effect of pin height H on pressure drop is more difficult to anticipate because the pin area increases but the flow area also increases. The experimental data largely follow these expected trends.

Figures 6-8 illustrate the behavior at zero tip clearance. Not surprisingly, Figs. 6 and 7 show that for fixed jet Re , the pressure drop through the heat sink is strongly linked to the pin density. Figure 6 compares the performance of increasingly dense heat sinks with identical pin diameter and Fig. 7 shows a 5x5 heat sink with three pin thicknesses. Additionally, the dependence of C_p on Re is weak below $Re = 10,000$ and the dependence is lost for $Re > 10,000$. This is not surprising since this behavior is commonly observed for fully turbulent internal flows. The data are not sufficiently refined to point out the laminar-turbulent transition but we may surmise that for $Re = 3000$ the flow may be laminar or transitional.

As discussed earlier, the effects of pin height on pressure drop are not easily predictable because the pins are exposed to flow that begins parallel to the pin axis and ends roughly normal to the pin axis. But the data of Fig. 8 show irrefutably that C_p decreases monotonically with increasing pin height, for fixed Reynolds number. If the flow remained parallel to the pin axis, the C_p no doubt would scale with the pin height, for fixed pin spacing, and the velocity parallel to the pin axis. On the other hand, for transverse flow, the pressure drop would scale with transverse

velocity and pin diameter, a . Defining Reynolds number based on the width, W makes no attempt to identify preferable definitions that acknowledge the two different types of flow, parallel and normal.

Figure 9 shows the pressure loss coefficient, C_p for heat sinks 1A, 1B and 1C against Re . Consistently with Fig. 8, as H/W decreases, C_p increases and becomes a weak function of Re .

As illustrated in Fig. 1, the clearance allows for an alternative flow path that bypasses the heat sink altogether. Prediction of the bypass is certainly not as straightforward as for side inlet/exist heat sinks [2]. Overall C_p measurements provide some opportunity for understanding overall effect of tip clearance. Figure 10 shows the pressure loss coefficient, C_p for the heat sink 1C, which has 8x8 configuration and a pin height of $H=22.5\text{mm}$, for various Reynolds number and clearance ratios of $h/H = 0, 1/8, 1/4$ and 1. As expected, the pressure loss coefficient C_p decreases as h/H increases due to the decrease in the overall hydraulic resistance as the bypass path gets larger. Note that C_p is a weak function of the Reynolds number, especially at low clearance ratio giving indirect evidence that the flow becomes fully turbulent. The dependence on Reynolds Number decreases with increasing h/H as well as with increasing Reynolds number. As mentioned in the earlier discussion, C_p goes to an asymptotic value of 1. When the clearance ratio, $h/H = 1$, the pressure loss coefficient, $C_p \cong 1$. This illustrates the fact that the pressure at the exit of the nozzle is atmospheric and that the flow through the heat sink is driven only by the initial momentum or dynamic pressure supplied to it and not the pressure drop across the heat sink. These behaviors are observed for all the heat sinks and are consistent for the entire set.

Figure 11 shows the effect of the pin height on the behavior of the pressure loss coefficient, C_p with bypass. In this plot, the densest heat sinks (8x8) are investigated and the pin diameter is kept fixed at $a = 1.5\text{mm}$. It can be seen that the pin height has a considerable effect on the performance of C_p since it is an important factor in determining the resistance opposing the jet flow. Consistently with what was expected, C_p diminishes as the pin height increases since more lateral flow area is provided for the air to pass through as it turns. In addition, the Reynolds number dependence is stronger for the short pinned heat sink, mainly at low clearance ratio, perhaps because the flow is not fully turbulent in this case. As h/H increases, this dependence diminishes and the pressure loss coefficient, C_p tends to 1 as previously observed. It is to be expected that such large clearance will also have detrimental effects on the heat transfer.

Figures 12 and 13 show the static pressure mapping along the transverse direction and the diagonal direction respectively. The results shown are for the densest heat sink (8x8), comparing two pin heights $H = 12.5\text{mm}$ and $H = 22.5\text{mm}$, which are the shortest and the tallest pin heights, respectively. It can be seen that the total pressure was almost 100% recovered at the center of the heat sink for the tall pins, whereas, only approximately 80% of this pressure is recovered for the short pins. The large loss in total pressure for the shorter heat sinks may be attributed to the severe turning radius experienced by the jet blowing into the heat sink. The small turning radius introduces large radial pressure gradients so that the static pressure at the center of the heat sink is lower than for cases in which the turning is larger. The static pressure drop is more severe for the taller heat sink in the transverse direction.

Close examination of Fig. 12 reveals that the static pressures are lower in the wall jet region outside of the short heat sinks than the taller heat sinks. This is probably due to the inviscid pressure recovery as a result of the flow deceleration, which is more pronounced in the tall heat sink. In the diagonal flow outside the heat sink, Figs. 13 and 15, demonstrate a mild pressure recovery just outside of the heat sink, at $D/(W/2) = 2$, especially noticeable for the sparse heat sink. This curious effect is most likely due to a low velocity region at the corner, observed for example by Danek and Moffat [8], in a square impinging jet. Because of the complexity of this flow, much interpretation remains to be performed and is the subject of continuing work.

Heat transfer

The primary experimental diagnostic for the heat transfer is the overall base-to-ambient thermal resistance, R , defined in Eq. 10. The heat transfer apparatus is designed to deliver roughly uniform temperature to the underside of the heat sink base, as shown in Fig. 5. The copper heat spreader uniformizes the base temperature, hence the temperature measured at the center of the heat sink base presumably represents the average base temperature. The magnitude of the temperature distribution within the pins in the transverse (thickness) direction is determined by the Biot number,

$$Bi = \frac{h^* \cdot a}{k} \quad (19)$$

For a heat transfer coefficient of order $100 \text{ W/m}^2 \cdot \text{K}$, and aluminum thermal conductivity of $151 \text{ W/m} \cdot \text{K}$, Bi is of order of 0.001. In the direction of the pin axis, the temperature distribution can be assessed by the fin efficiency, defined as [15]:

$$h_f = \frac{Q_f}{Q_{\max}} = \frac{Q_f}{h^* A_f \Delta T_B} \quad (20)$$

For the pins of square cross-section of length H , neglecting tip heat loss:

$$h_f = \frac{\tanh(m \cdot H)}{(m \cdot H)} \quad (21)$$

where

$$m = \sqrt{\frac{h^* P^*}{k A_c}} = \sqrt{\frac{4h^*}{ka}} \quad (22)$$

The fin efficiency varies from 0.94 for short, thick pins to 0.75 for slender tall pins used in this study. Hence, for the majority of the heat sinks, it cannot be assumed that the pins are isothermal. The pin temperature varies along its length, and in addition, the average pin temperature depends on location in the array because of the spatial variability of the heat transfer coefficient. It is useful to keep in mind therefore that the resistance R at best is useful for understanding the parametric performance of heat sinks, but is not very helpful for understanding the flows and the attendant variations of heat transfer.

Results with no bypass are examined first. Figures 16 and 17 demonstrate that the overall thermal resistance decreases monotonically with Reynolds number for all pin densities, Fig. 16, and pin diameters, Fig. 17. These figures show data for the tallest heat sinks, $H/W = 0.9$, but similar results are observed at all pin heights. As the number of pins increases, surface area increases and flow area decreases. Not surprisingly, the heat sink resistance decreases with number of pins (Fig. 16), and the pressure loss coefficient increases. Similarly, as the pin diameter increases, the fin efficiency increases, thereby decreasing the overall thermal resistance.

These effects can readily be seen by recalling the definition of an overall surface efficiency [15]:

$$h_o = \frac{Q_B}{Q_{\max}} = \frac{Q_B}{h^* A_T \Delta T_B} \quad (23)$$

where the total area is the sum of the pin area ($N \times A_{pin}$) and the unpinned base area, the so-called “prime area”, A_{prime} :

$$A_T = N \times A_{pin} + A_{prime} \quad (24)$$

It is readily shown that:

$$h_o = 1 - \frac{N \cdot A_{pin}}{A_T} (1 - h_f) \quad (25)$$

and from the definition of base-to-ambient resistance:

$$R = \frac{1}{h_o h^* A_T} \quad (26)$$

In these definitions, it is assumed that h^* is uniform over all surfaces. The overall thermal resistance of

Fig.’s 18 and 19 reveal a remarkable observation about the dependence of overall thermal resistance on pin height, that points out the fallacy of assuming that the heat transfer coefficient is constant over all surface areas. Figure 18 shows the dependence of R on Re for an 8x8 array at three different heights. At the lowest Re , R decreases as the height increases. This is seen more clearly in Fig. 19. However, the decrease in R in varying H/W from 0.5 to 0.9 is far less than what would have been anticipated based on surface area change alone, i.e. assuming that h^* remains constant. It is even more remarkable to observe that at the highest Reynolds number, $Re=30,600$, there is no significant dependence of thermal resistance on pin height, despite the fact that the pin surface area nearly doubles. In contrast, Fig. 8 shows that C_p drops from 3.2 to 1.8, a factor of nearly two as the pin height changes from $H/W = 0.5$ to 0.9, for the 8x8 array, at the highest Reynolds number. Similar observations can be made for the other pin densities. Two important conclusions can be drawn from the discussion of Fig.’s 8 and 19. First, there appears to be very little thermal performance benefit in increasing the pin height from $H/W = 0.5$ to 0.9, remarkable but irrefutable. However, the reduction in pressure drop as a result of the increased lateral flow area is significant, and this would be the primary benefit in the design of a flow system to deliver flow to impinge on heat sink. Secondly, the insensitivity of overall R to pin height H/W for fixed jet Reynolds number, requires that there be compensating mechanisms. These are illustrated by Eq.’s 24 and 26. As pin height increases, A_T increases because $N \times A_{pin}$ increases. However, as the flow turns to exit the lateral faces of the heat sink, increased flow area is encountered, allowing deceleration of the cross-flow velocity component, and presumably a reduction of the per pin heat transfer coefficient. A second mechanism may also be hypothesized. For small H/W , most of the cooling flow is in cross-flow to the pins because the flow must turn abruptly, compared to taller pins. The heat transfer coefficient per unit pin length is high in these situations. For taller pins, the increased lateral flow area may allow a less aggressive turning towards the lateral faces, thereby exposing less of the pin area to cross-flow, and more of it to in-line flow. The net result is that the per pin heat transfer coefficient per unit length decreases as pin height increases.

Effect of Flow Bypass

The strong dependence of C_p on tip clearance is illustrated in Fig. 11. It is surprising that the thermal resistance has little dependence on tip leakage at elevated Reynolds number, and mild dependence at

low Reynolds number. These trends are observed in Fig.'s 20 and 21. The thermal resistance increases by a maximum of 20% at a clearance ratio of unity, at lowest Re as shown Fig. 20. Sensitivity to tip clearance is highest for the shortest heat sinks, $H/W = 0.5$, and at the lowest Reynolds number. This is true for both C_p and R . Indeed at high Reynolds number, the thermal resistance, R is insensitive to tip clearance, at all pin heights. Thus by comparison of Fig.'s 11 and 21, it is of interest to note that a 70% decrease in pressure drop can be accomplished with a 25% tip clearance ratio, with no penalty on the thermal performance for $H/W = 0.9$. The gains in pressure drop are even greater for shorter heat sinks, but a larger penalty on thermal performance is evident.

CONCLUSIONS

1. For all of the heat sinks tested, the pressure loss coefficient showed little dependence on Reynolds number. This may be because the flow is in a fully turbulent regime.
2. At any jet Reynolds number, pressure drop increased with pin density and pin diameter, and decreased with increasing pin height.
3. Pressure drop decreased strongly with increasing tip clearance, with most of the reduction occurring within clearance ratios up to 25%. The effects of tip clearance were most pronounced for smaller pin heights.
4. Variations of the local static pressure within the heat sinks were very complex. The stagnation pressure recovery at the center of the heat sink surprisingly was higher for tall pins than for short pins.
5. The overall thermal resistance decreased with increasing jet Reynolds number, with increasing pin density, and with increasing pin diameter.
6. Remarkably, the thermal resistance was weakly dependent on pin height. It is hypothesized to be due to the fact that the heat transfer coefficients are much higher in cross-flow than parallel to the pin axis.
7. At high Re , overall thermal resistance was not strongly sensitive to tip clearance ratio. Short pins at low Re were most strongly sensitive to variations in tip clearance.

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Table 1 Geometric description of heat sinks tested.
All dimensions in mm.

Desc.	a	b	H	n	a/H	b/a	H/W
1A	1.5	1.86	12.5	8	0.12	1.24	0.5
1B	1.5	1.86	17.5	8	0.09	1.24	0.7
1C	1.5	1.86	22.5	8	0.07	1.24	0.9
2A	1.5	2.42	12.5	7	0.12	1.61	0.5
2B	1.5	2.42	17.5	7	0.09	1.61	0.7
2C	1.5	2.42	22.5	7	0.07	1.61	0.9
3A	1.5	3.2	12.5	6	0.12	2.13	0.5
3B	1.5	3.2	17.5	6	0.09	2.13	0.7
3C	1.5	3.2	22.5	6	0.07	2.13	0.9
4A	1.5	4.38	12.5	5	0.12	2.92	0.5
4B	1.5	4.38	17.5	5	0.09	2.92	0.7
4C	1.5	4.38	22.5	5	0.07	2.92	0.9
5A	1.5	6.33	12.5	4	0.12	4.22	0.5
5B	1.5	6.33	17.5	4	0.09	4.22	0.7
6A	2	3.75	12.5	5	0.16	1.88	0.5
6B	2	3.75	17.5	5	0.11	1.88	0.7
6C	2	3.75	22.5	5	0.09	1.88	0.9
7A	2.5	3.13	12.5	5	0.20	1.25	0.5
7B	2.5	3.13	17.5	5	0.14	1.25	0.7
7C	2.5	3.13	22.5	5	0.11	1.25	0.9

Table 2 Description of instruments Uncertainties.

Instrument	Description
Pressure Transducer	Accuracy: +/- 0.12% of reading
Digital Readout	Accuracy: +/- 0.01% of reading
A/D Converter	12 bit, Span: +/- 10 V

Table 3 Estimated Inviscid Pressure Recovery for Impinging Jet.

H	H/W	A_n / A_{ex}	$C_{p_{area}}$
12.5	0.5	0.5	0.25
17.5	0.7	0.36	0.1296
22.5	0.9	0.28	0.0784

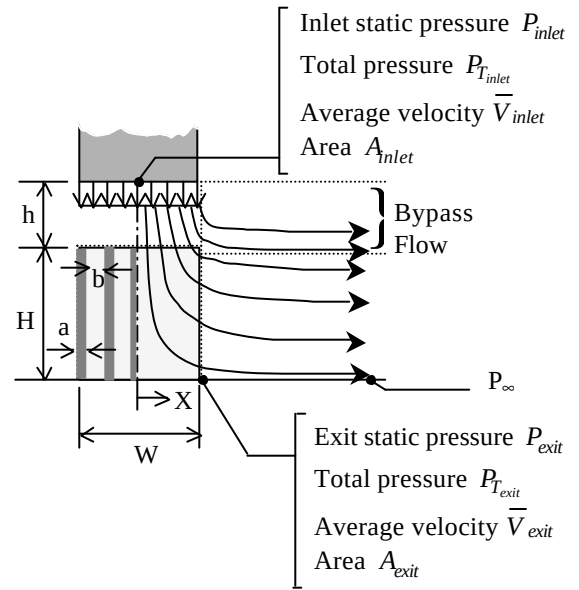


Figure 1 Flow behavior in jet impingement on pins with tip clearance.

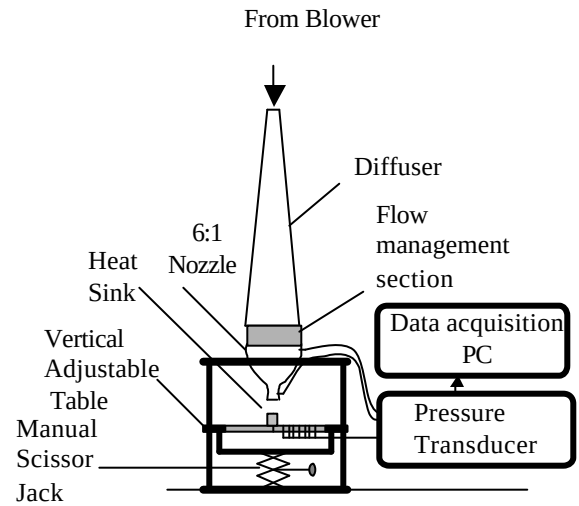


Figure 2 Schematic of the experimental set-up.

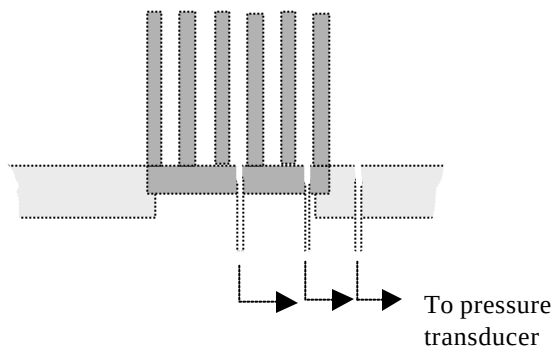


Figure 3a Schematic of the hydraulics test plate, side view.

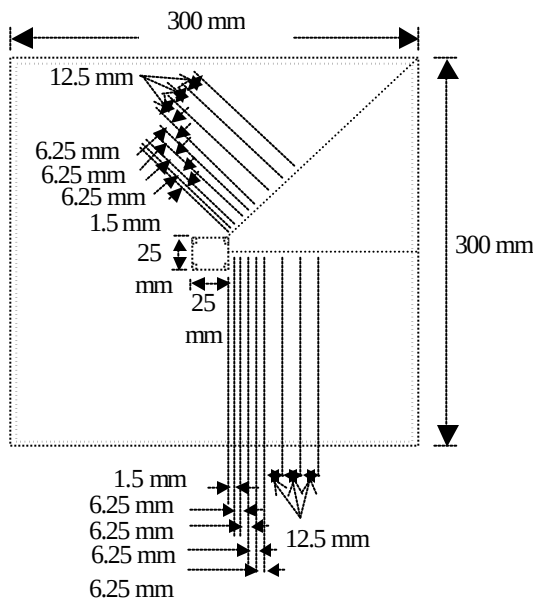


Figure 3b Schematic of the hydraulics test plate, top view.

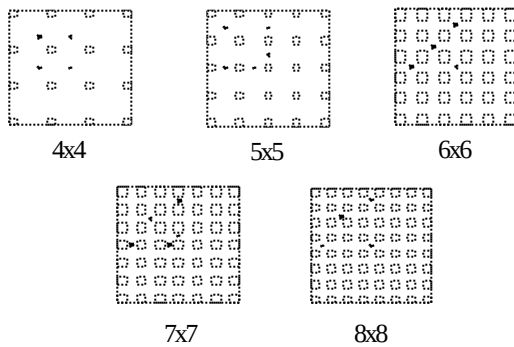


Figure 4 Illustration of the pressure tap locations for the various heat sink families.

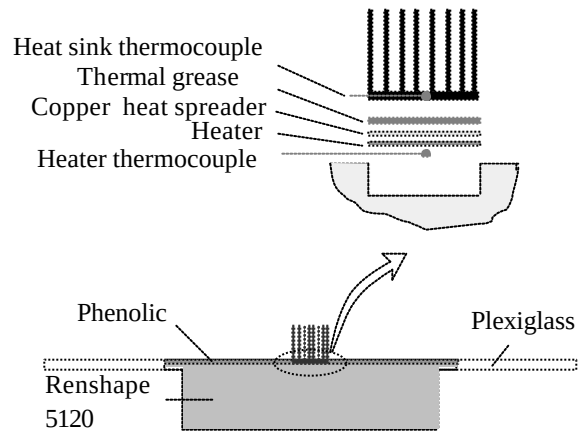


Figure 5 Schematic of the heat transfer plate.

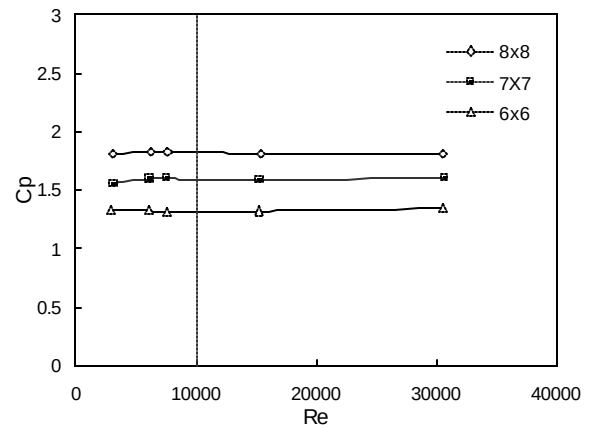


Figure 6 Pressure loss coefficient at fixed pin height $H/W = 0.9$ and $h/H = 0$ for different pin arrays with $a/H = 0.12$.

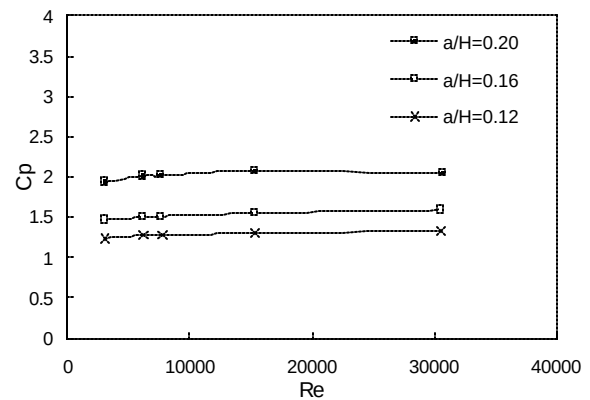


Figure 7 Pressure loss coefficient at fixed pin height $H/W = 0.9$ and $h/H = 0$ for heat sinks 4C, 6C and 7C with varying pin thickness.

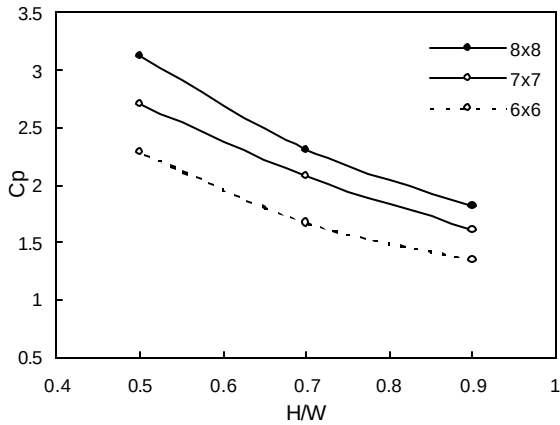


Figure 8 Pressure loss coefficient vs. aspect ratio (H/W) for three different pin densities at $Re = 30600$, $h/H = 0$.

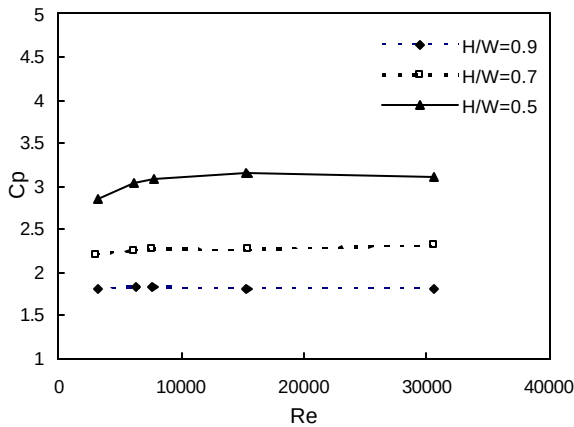


Figure 9 Dependence of pressure loss coefficient on Reynolds number for heat sinks of different heights 1A, 1B and 1C (8x8) at $h/H = 0$.

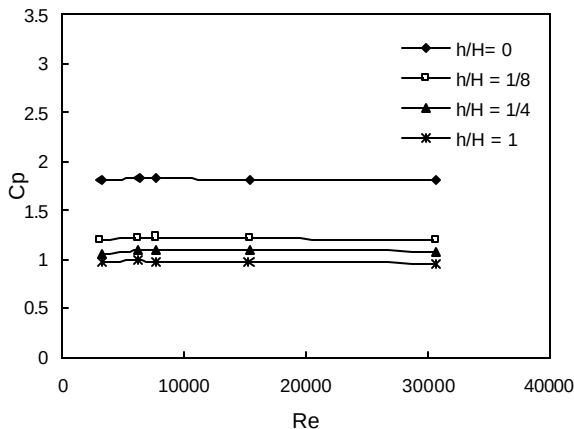


Figure 10 Pressure loss coefficient for 1C ($H/W = 0.9$) at various clearance ratios.

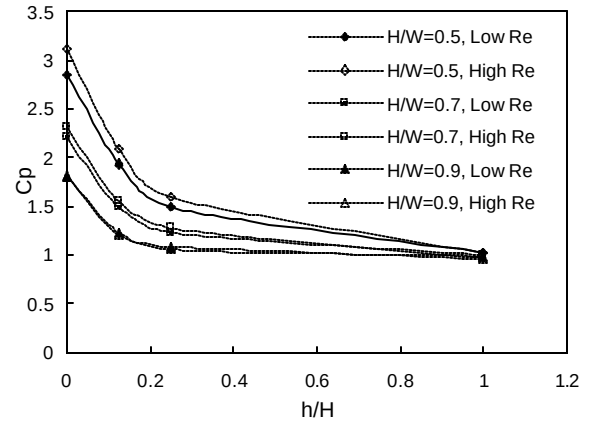


Figure 11 Effect of tip clearance on overall pressure loss coefficient for 1A, 1B and 1C (8x8) at various Re .

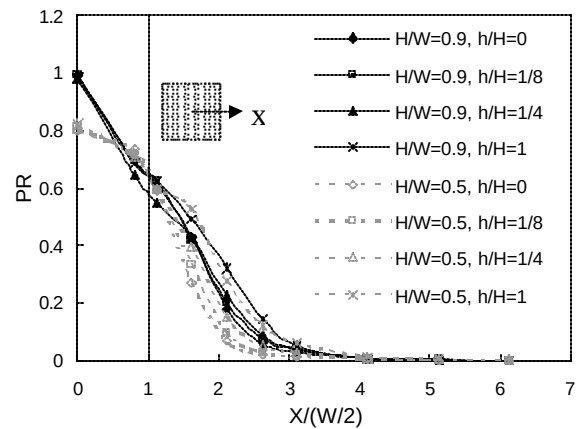


Figure 12 Pressure mapping for 1A and 1C (8x8) at different clearance ratios along the lateral direction of the heat sink at $Re = 7650$.

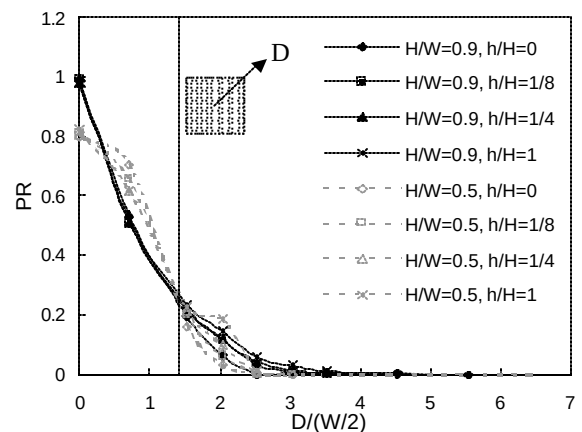


Figure 13 Pressure mapping for 1A and 1C (8x8) at different clearance ratios along the diagonal direction of the heat sink at $Re = 7650$.

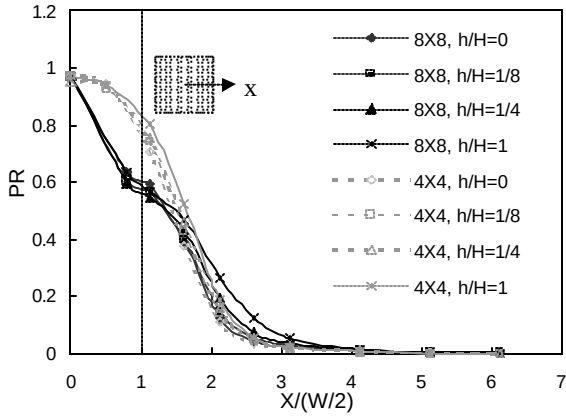


Figure 14 Pressure mapping for 1B and 5B at different clearance ratios along the lateral direction of the heat sink at $Re = 7650$ and $H/W = 0.7$.

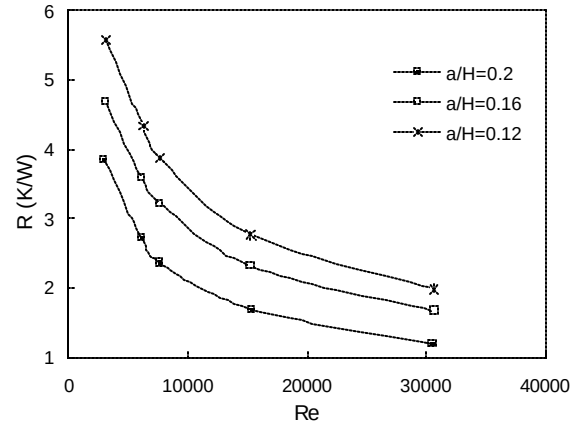


Figure 17 Heat sink base-to-ambient resistance at fixed pin height $H/W = 0.9$ and $h/H = 0$ for heat sinks (5x5 arrays) with varying pin thickness.

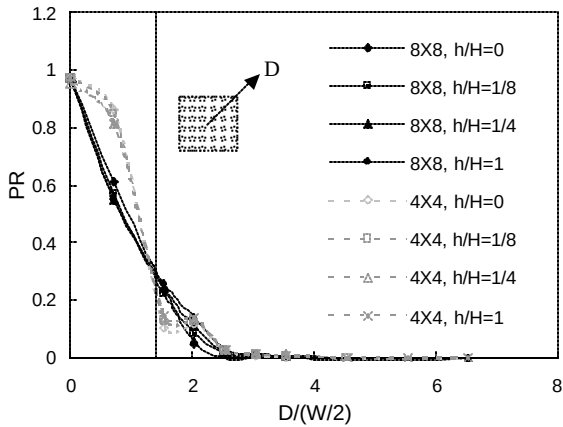


Figure 15 Pressure mapping for 1B and 5B at different clearance ratios along the diagonal direction of the heat sink at $Re = 7650$ and $H/W = 0.7$.

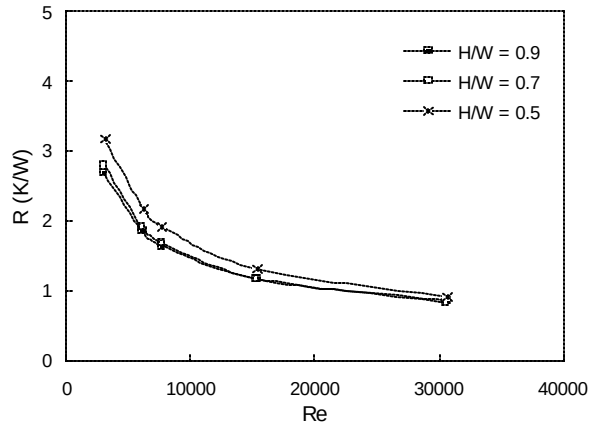


Figure 18 Heat sink base-to-ambient resistance vs. Re number at $h/H = 0$, for 1A, 1B and 1C (8x8).

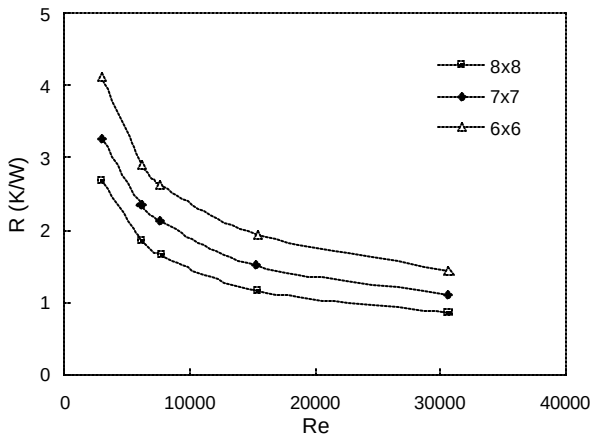


Figure 16 Heat sink base-to-ambient resistance dependence on Reynolds number and pin density for $a/H = 0.12$ at fixed pin height $H/W = 0.9$ and $h/H = 0$.

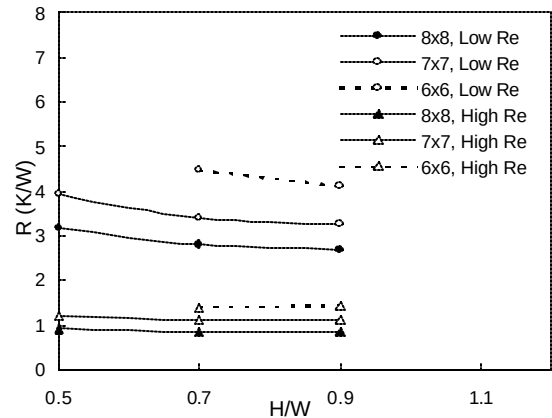


Figure 19 Heat sink base-to-ambient resistance vs. aspect ratio at High $Re = 30600$ and Low $Re = 3000$ $h/H = 0$ for three different pin densities.

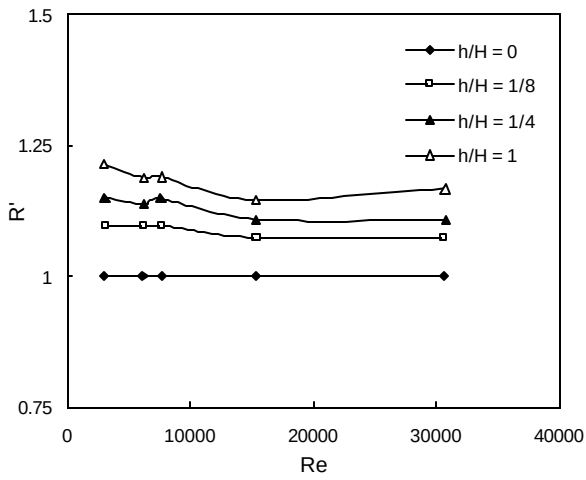


Figure 20 Normalized heat sink base-to-ambient resistance vs. Re for 1C at different clearance ratios.

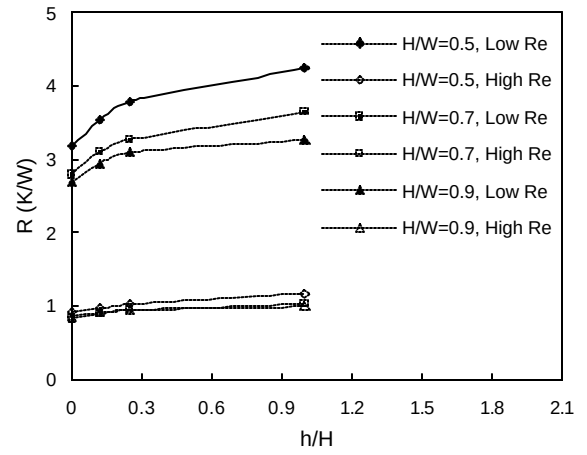


Figure 21 Overall thermal resistance vs. clearance ratio at high Re (30600) and low Re (3000) for 8x8 heat sink.